# Zaid Ahmad

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## Education

**Carleton University** Ottawa, ON

*B.Sc. Computer Science, AI and Machine Learning Stream, Minor in Mathematics* *Sept 2024 - May 2029*

- **Coursework:** Data Structures and Algorithms, C/C++, Linear Algebra, Discrete Structures II, Web Applications

## Technical Skills and Clearances

**Languages:** Python, SQL, TypeScript, JavaScript, C++, C, Java, Go
**Databases:** PostgreSQL, SQLite, MongoDB, Firebase, Supabase, JSONB, Schema Design, Indexing
**Frameworks:** FastAPI, Node.js, Express, React (Next.js), React Flow, Tailwind CSS, Hugo, Parsel, BeautifulSoup, scikit-learn
**Cloud and Tools:** AWS, Azure DevOps, Railway, Cloudflare, Neon, Vercel, Docker, GitHub Actions, CI/CD, Git, Linux
**Concepts:** REST APIs, Unit Testing (Vitest, pytest), Machine Learning, Networking, Concurrent and Systems Programming
**Clearance:** Government of Canada Reliability Status Security Clearance, Issued 2026

## Experience

**Business Analyst (FSWEP)** June 2026 - Aug 2026

*Immigration, Refugees and Citizenship Canada* *Ottawa, ON*

- Intake spanned case-management and non-case-management requests across four categories. Triaged submissions in Azure DevOps and classified scope and category (Cloud, AI, Software, Potential Project), standardizing dashboard entries covering business need, impacts, and consultation.
- Business owners submitted generic, unquantified impact claims. Issued targeted clarification correspondence and flagged unsupported claims as follow-up gaps, converting input into quantified statements on transaction volumes, affected client records, and multi-system scope.
- Approved business needs had to reach review as auditable requirements. Authored High-Level Business Requirements, including for a governed low-code application platform, self-auditing each draft into numbered capability statements traceable to source.
- Recurring annual change requests returned every governance cycle. Updated precedent figures, dates, and references instead of redrafting, routed qualifying items to an expedited path, and reconciled the departmental prioritization list after each review, cutting drafting turnaround.
- Drafting conventions lived in individual practice, not in writing. Codified business need structure, impact standards, official-languages requirements, and a pre-send checklist into a shared reference, improving consistency of team intake output.

**Software Developer (Volunteer)** May 2026 - Aug 2026

*Carleton Computer Science Society* *Remote*

- The course-graph Explorer had no way to find a course without scanning the graph. Implemented a type-ahead search in React and TypeScript with normalized code and title matching, reusing the existing store so results highlight identically to a node click.
- The codebase ran tests without reporting coverage. Integrated @vitest/coverage-v8 and configured HTML and terminal reports in vite.config.ts, landing through every lint, format, and typecheck CI gate.
- Explorer users could not narrow the graph to core courses. Implemented a React and TypeScript toggle deriving the visible node set outside the layout engine and recomputing dagre layout on state change, establishing a reusable filter seam.

**Logistics Lead** Sept 2025 - Present

*cuHacking* *Ottawa, ON*

- A 500+ attendee hackathon needed venue, equipment, and day-of coordination. Led a 2-person logistics team across setup and live operations.

## Projects

**CarletonCourseMap** carletoncoursemap.ca/map | github.com/zaidahmad16/CarletonCourseMap

*Python, Parsel/XPath, FastAPI, PostgreSQL, Next.js, React Flow, dagre, Tailwind CSS, Railway, Cloudflare*

- Carleton students planned degrees by reading prerequisite text course by course. Scraped Carleton Central with Parsel/XPath, modelled AND/OR prerequisite logic as a DAG, and stored course data as JSONB in PostgreSQL on Neon, eliminating manual planning for 1,700+ students.
- Scraped course, program, department, prerequisite, and instructor data had to stay consistent. Designed a normalized PostgreSQL schema with foreign key constraints holding 3,218 courses, 304 programs, and 8,882 prerequisite edges, answering every API query in under 2 ms on the server.
- Prerequisite conflicts across a full degree plan were invisible in a course list. Rendered four-year dependency graphs in React Flow and dagre with color-coded nodes and monthly-refreshed RateMyProfessors ratings, surfacing conflicts in real time.
- Serving the graph publicly exposed the data to open traffic. Built a FastAPI backend with rate-limited, CORS-protected endpoints, batch lookup of 100 records per round-trip, and API-key-gated admin routes for curating graph layouts.

**OntarioDriveTestMap** ontariodrivetestmap.fyi | github.com/zaidahmad16/OntarioDriveTestMap

*Python, PaddleOCR, pyosmium, SQLite, HDBSCAN, OSRM, FastAPI, PostgreSQL, React, Leaflet, Railway, Cloudflare*

- Ontario does not publish road test routes, and competing sites sell sample maps. Reconstructed routes for 3 test centres from Reddit posts and dashcam video, shipping 13 routes with per-segment confidence labels.
- Crowd and video evidence had to stay queryable as routes were rebuilt. Designed a PostgreSQL schema linking 56 source traces, their waypoints and turns to 101 consensus segments and 13 routes, serving each centre's full map in about 1 ms of query time.
- OCR on dashcam footage reported zero readable frames. Traced it to a swallowed PaddleOCR crash and destructive downscaling, then reran at 1080p on a Colab GPU, raising street reads on one video from 2 to 114.
- Nominatim could not resolve intersections. Built an offline index of 19,482 OpenStreetMap junctions with pyosmium and joined unnamed highway ramps with union-find, resolving highway accounts to real roads.
- Crowd accounts were partial and often repeated one author. Scored segments with author deduplication, source weighting, and age decay, clustered with HDBSCAN, and found by leave-one-out validation that the planned threshold cut F1 from 0.79 to 0.64.